

A vector logic for formal semantics

Formal semantics interprets the expressions of each type in a typed model (Montague, 1974). Distributional semantics and neural language models represent expressions as vectors in a geometry learned from use. I study which parts of a typed semantics admit an exact encoding in vector spaces, and what form semantic computation takes where linearity fails. The answers are theorems; under bridging hypotheses, they yield predictions about learned representations and about speakers.

The base theorem (Quigley, 2025a) grew out of my dissertation (Quigley, 2025b): it assigns to each typed domain $\mathcal{D}_\tau$ a free carrier $\mathcal{F}_\tau$, the real vector space with a basis vector $\mathbf{b}_a$ for each $a \in \mathcal{D}_\tau$, together with the injection $h_\tau : a \mapsto \mathbf{b}_a$. Truth values go to an orthonormal pair $\mathbf{b}_1, \mathbf{b}_0$ in $\mathbb{R}^2$, and an n -ary connective goes to a 2×2^n matrix on the n -fold tensor power of $\mathbb{R}^2$ (Mizraji (1992) introduced this matrix calculus for propositional logic). We extend it to typed λ -terms: every semantic function f lifts to a unique multilinear map L_f , and the lift of a composite is the composite of the lifts. For unary $f : \mathcal{D}_\sigma \rightarrow \mathcal{D}_\tau$, the square

$$\begin{array}{ccc} \mathcal{D}_\sigma & \xrightarrow{f} & \mathcal{D}_\tau \\ h_\sigma \downarrow & & \downarrow h_\tau \\ \mathcal{F}_\sigma & \xrightarrow{L_f} & \mathcal{F}_\tau \end{array}$$

commutes. The compact-closed semantics of Coecke et al. (2010) works in finite dimension, and its pregroup grammar relates two types by at most one reduction; the free construction holds at every cardinality, and distinct semantic functions receive distinct lifts.

Each development of the encoding recovers the layer beneath it, as a quotient or a special case (see Figure 1); the recovery is a theorem in each case. The extensional paper is published and the intensional paper is under review at the same journal; the remaining material is in development.

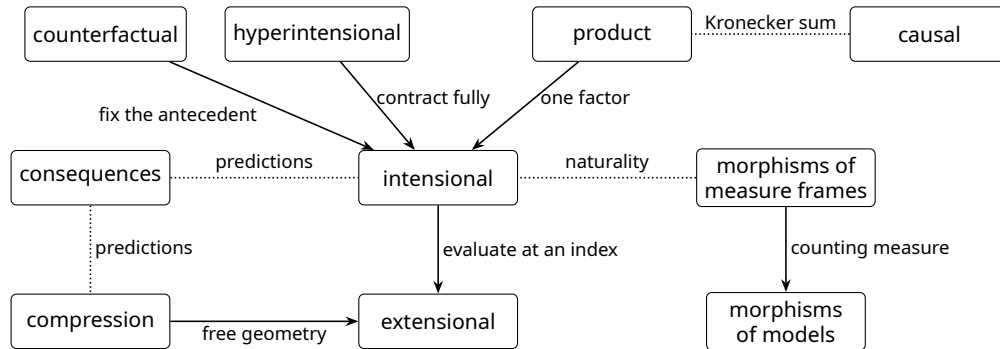


Figure 1: A solid arrow runs from a paper to the theory it recovers, under the labeled restriction. Dotted lines mark dependencies of other kinds: predictions drawn from the encoding, naturality of the encoding along morphisms, and the failure of the light cone to factor as a Kronecker sum. The causal layer applies the operators of the intensional and product layers to causal accessibility.

Descent

The intensional layer adds a finite set of index sorts (worlds and times among them), and gives each type a second carrier beside the free one (Quigley, 2026c). The operator carrier of a function type $\langle \alpha, \beta \rangle$ is the space $\text{Hom}(\mathcal{F}_\alpha, \mathcal{S}_\beta)$ of linear maps, where an intension is an operator on the free carrier of the index space.

A Boolean functional F on a function domain descends when some linear map Λ on the operator carrier satisfies $\Lambda(h(f)) = h_t(F(f))$ for every argument f . A Boolean-valued functional on a Boolean algebra of sets descends if and only if it is constant, the indicator of an ultrafilter, or the complement of such an indicator. On a finite domain, the nonconstant descending functionals are of the type of generalized quantifiers: Montague’s individuals $\lambda P. P(d)$ and their negations. Grefenstette (2013) showed, by scaling an argument, that universal and existential quantification fail to be multilinear on set vectors; the descent theorem extends this failure to every linear map on the operator carrier. Once the restrictor has two or more elements, the scope functionals of *every*, *some*, *no*, *most*, and the numerals up to its size fail to descend. A necessity operator at an index descends if and only if its accessible set is null or an atom of the measure, which on a discrete frame means at most one accessible index. Belief, as necessity over doxastic alternatives, descends only for an agent with, at most, one alternative, up to null sets.

Outside the linear regime, these operators compute by a linear accumulation against a kernel and a decision. For a discrete frame with accessibility matrix $\mathbf{A}$ and a proposition with truth profile $\mathbf{v}(\varphi) \in \{0, 1\}^n$, the modal clauses read

$$(\Box\varphi)(w_i) = 1 \iff (\mathbf{A} \mathbf{v}(\varphi))_i = \deg_{\mathbf{A}}(w_i), \quad (\Diamond\varphi)(w_i) = 1 \iff (\mathbf{A} \mathbf{v}(\varphi))_i > 0,$$

where $\deg_{\mathbf{A}}(w_i)$ is the out-degree of w_i . By the tree of numbers of van Benthem (1986), a conservative, extension-invariant, and isomorphism-invariant determiner is a decision on the pair $(|P \setminus Q|, |P \cap Q|)$. Both counts are bilinear accumulations on encoded predicates.

I state modality uniformly over measure frames, which equip each index with a measure on its accessible set. Under counting measure, the clauses are those of Kripke semantics at every cardinality, and under a continuous measure, necessity becomes truth-almost-everywhere on the accessible set. Normalized by the mass of the accessible set, the accumulation is a conditional probability; thresholds on it give the probability operators of graded modality. The almost-everywhere box is the threshold-one member of the family, and its only nontrivial normal member.

Morphisms

In Quigley (2026a), I determine which maps between typed models preserve denotation. A map on entities and on each index frame induces a logical relation at every type; a morphism is such a map under which each lexical item is related to its counterpart, and each frame map is a bounded morphism. By the fundamental lemma of logical relations (Statman, 1985), every expression then denotes related values in the two models. The modal case follows: the necessity operator is related to its counterpart precisely when the frame map is a bounded morphism.

For standardly interpreted items, relatedness becomes a condition on the map: on entities, identity forces injectivity; *every*, *some*, and *no* force surjectivity; *most* forces surjectivity with fibers of equal size on finite domains; the counting determiners force bijectivity. On index frames, given boundedness, a counting modality forces injectivity on each accessible set, and a majority modality bounds the largest fibers over it. A lexicon with a counting determiner, thus, makes every morphism bijective on entities. With an empty set of index sorts, the models of such a lexicon form a groupoid.

Logical relations can fail to compose: at the type $\langle\langle\sigma, \sigma\rangle, t\rangle$, the composite of two morphisms can violate lexical relatedness. On composable types, which include the polarized types, models and morphisms form a category. On polarized types, the carriers and the encoding of the intensional layer are functorial along morphisms.

Compression

On one-hot inputs, an embedding layer with matrix $\mathbf{E} \in \mathbb{R}^{V \times d}$ is the linear map $\mathbf{H} = \mathbf{E}^\top$ on the free entity carrier (Quigley, 2026b). I call such a geometry “free” when its entity vectors are linearly independent, and “compressed” otherwise. For a monadic lexicon with truth matrix $\mathbf{T}$, exact linear decoders into a shared truth basis exist if and only if the row space of $\mathbf{H}$ contains that of $[\mathbf{T}; \mathbf{1}]$, the truth matrix augmented by the all-ones row. The least dimension of such a geometry is $\text{rk}[\mathbf{T}; \mathbf{1}]$, and the least dimension for affine decoders is one lower.

A lexicon closed under all predicates forces the free regime. Once the atoms decode exactly, the connective matrices act on their outputs unchanged, and determiners compute by two bilinear accumulations through the Gram matrix $\mathbf{H}\mathbf{H}^\top$. For a binary relation R , an exact bilinear decoder exists if and only if $\mathbf{1}$ lies in the row space of $\mathbf{H}$ and the truth matrix factors as $\mathbf{T}_R = \mathbf{H}^\top \mathbf{A}_R \mathbf{H}$. Identity and every strict total order, therefore, force the free regime.

The defect of a predicate is the least-squares distance from its truth row to the row space of $\mathbf{H}$. A linear probe (Alain and Bengio, 2017) tests separability, a strictly weaker property than exact decoding. Over GloVe (Pennington et al., 2014) and word2vec (Mikolov et al., 2013), against the McRae feature norms (McRae et al., 2005) and WordNet hypernyms (Fellbaum, 1998), every predicate has positive affine defect. Strict separability is common nonetheless: on WordNet, 159 of 171 predicates are separable over GloVe, against 87 over a Gaussian control of the same shape.

Trained transductively under the truth term of a mixed objective, a geometry reaches exact affine decoding to numerical precision at every tested dimension from $\text{rk}[\mathbf{T}; \mathbf{1}] - 1$ upward. At the embeddings' dimension of 300, a geometry constrained to exact linear decoding retains 98.5 to 99.2 percent of the pretrained variance on the feature norms and 79.5 to 82.5 percent on WordNet. Dimension is, thus, excluded as the reason the pretrained embeddings miss the criterion; whether exactness generalizes to unseen entities is current work.

Consequences

The encoding theorems hold of every typed model, and the rank criterion holds of every geometry; their empirical content lies in bridging hypotheses, which state how speakers and learned systems realize the objects of the semantics.

A Boolean function of two arguments that is a sum $f(x) + g(y)$ depends on one argument only. An affine decoder applied to separately computed representations of restrictor and scope thus decides a function of one of them, whatever the representations; deciding a determiner requires an interaction of the two. In the algebra, that interaction is a bilinear accumulation. For transformers, I predict that the truth of *every P is Q* becomes linearly decodable at the sublayer, where attention aggregates the restrictor; the prediction fails if truth decodes linearly at a layer where every accumulation is undecodable. Below exactness, the compressed count of $P \cap Q$ errs by exactly the inner product of the least-squares residuals of P and Q . A decision on compressed counts should then agree with the true decision wherever its margin exceeds that error.

Lidz et al. (2011) report that speakers verify *most* by comparing $|P \cap Q|$ with $|P| - |P \cap Q|$. Affine decoders of atoms reach $|P \setminus Q|$ only by that subtraction, because the truth row of $P \wedge \neg Q$ lies outside the affine span of the atoms when P and Q vary independently. Under their interface transparency hypothesis, the subtraction should also appear in the verification of *every* and *fewer than half*.

If category judgments are affine functions on the space that supports similarity judgments, then the dimension d_B of that space is at least $\text{rk}[\mathbf{T}; \mathbf{1}] - 1$ for every lexicon judged exactly. For the McRae lexicon, the bound is 76. A sparse embedding of the THINGS odd-one-out judgments has 49 dimensions (Hebart et al., 2020), and 66 with more data (Hebart et al., 2023); those numbers come from a sparsity-penalized model on a different concept set. Measuring both quantities on the concepts THINGS shares with a feature-norm database makes the comparison direct.

For the almost-everywhere box, an exception a speaker can report occupies an interval or region of positive measure; on every reportable state, it agrees with the strict box. Extended to generics, the threshold family of graded operators is refuted: speakers accept *mosquitoes carry West Nile virus* at a prevalence near one percent and reject *books are paperbacks* at a prevalence above one half (Leslie, 2008).

Sketch of future development

Over measure frames, each modal operator determines the class of measurable maps that preserve it. The class for the accumulation is strictly smaller than the class for necessity. The linear stage of a modal operator is thus a finer invariant than its decision. Strong *must* and *probably* have incomparable invariance classes. The

kernel form also turns the correspondence conditions for 4 and 5, which the intensional paper proves in one direction, into biconditionals.

The counterfactual layer replaces the fixed accessibility matrix with a family $\{S_A\}$ of selection matrices indexed by antecedents. For each fixed antecedent, the conditional is a normal box against S_A . Variable strictness enters because S_A depends on A through a selection outside the bilinear maps. On finite, totally ranked, strongly centered vector models, the valid formulas are those of VC (Lewis, 1973), with completeness transferred from the sphere semantics of Lewis.

In the hyperintensional layer, evaluation is the contraction of a network of operator matrices and argument vectors. The hyperintension of a formula is its decorated derivation, encoded faithfully outside the image of the truth embedding. On finite index sets, full contraction recovers intensional evaluation exactly. Finer grains support operators that provably fail to factor through the intension. For belief, the failure of logical omniscience is a theorem.

The product layer admits several index sorts at once. Since $\Box_M = \Box_{\text{supp } M}$ and $\Box_{MN} = \Box_M \Box_N$ for nonnegative matrices, the operator map factors through the Boolean semiring. The boxes of M and N commute precisely when the Boolean supports of MN and NM agree, a strictly weaker condition than $MN = NM$. Every frame-level theorem of the layer is inherited from many-dimensional modal logic (Gabbay et al., 2003), within the finite products.

On finite causal sets (Bombelli et al., 1987), the logic of order intervals under the matrix modality is exactly Grz.2; it extends by Grzegorczyk's axiom the logic S4.2 of causal accessibility on Minkowski spacetime (Goldblatt, 1980). On Minkowski spacetime of dimension at least two, the almost-everywhere modality assigns causal and chronological accessibility one operator, since the two relations differ by a null set at each point. Its logic strictly contains the chronological logic OI of Shapirovsky and Shehtman (2003). I conjecture that OI with boxed reflexivity $\Box(\Box p \rightarrow p)$, a sound Sahlqvist axiomatization, is complete for it. The light cone fails to factor as a Kronecker sum of a temporal and a spatial relation, while its coordinatewise sections factor.

Agenda

The count-error prediction and the defect profile of the THINGS embedding are computable now from published data with the existing pipeline. I plan to test the transformer prediction by probing open-weight models for the accumulations of quantified sentences. Replacing the pretrained matrix in the mixed objective with corpus statistics tests whether exact truth-conditional structure can emerge from co-occurrence, and at what distributional cost. I expect exactness to require the truth term.

On the formal side, the completeness conjecture for the almost-everywhere causal logic reduces to the finite model property of the axiomatization and to a measure p -morphism from the plane onto each of its finite rooted frames. Whether composability of lexical types is necessary for morphisms to compose is open, and so is the question of which classes of frame maps arise from finite lexicons.

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